

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

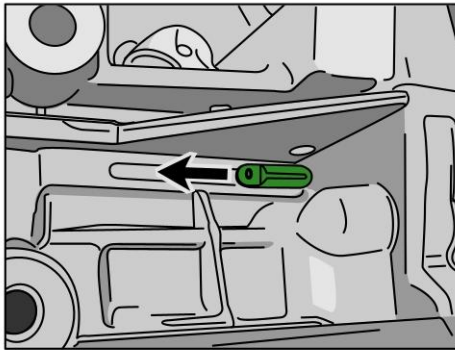
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

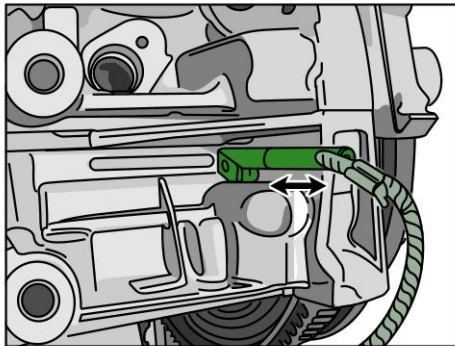
Always use new gaskets

Checking

Remove the access plug



Fit the crankshaft locking tool (11 0 300)

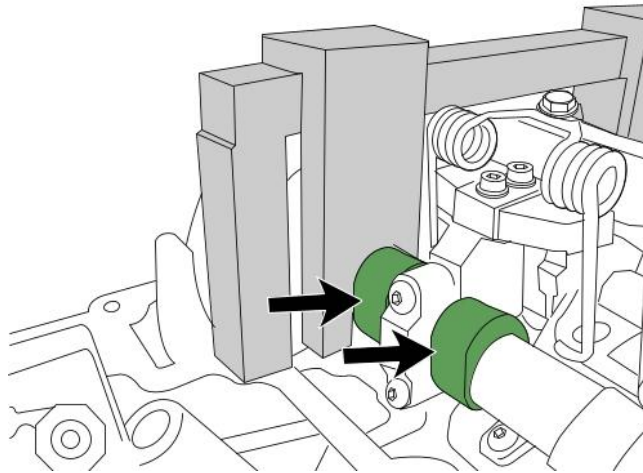


Turn the crankshaft until it touches the TDC locking pin

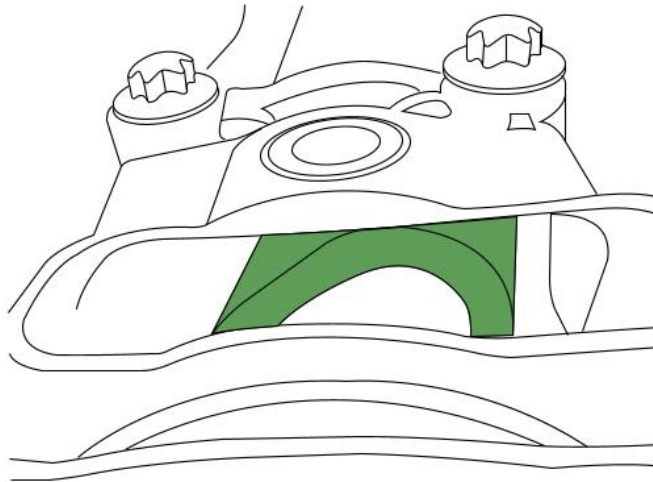
The crankshaft should be at TDC for cylinder 1

Note: Both camshafts must point upwards at an angle

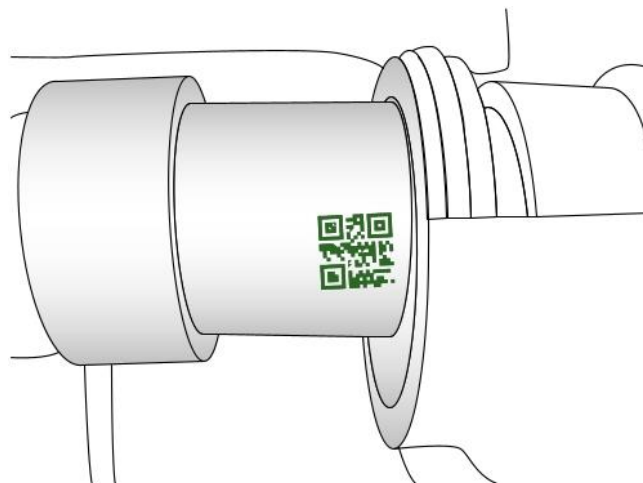
Inlet camshaft cam angle

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Exhaust camshaft cam angle



Note: Ensure that the marks face upwards

**Removal**

Disconnect the battery

Remove the wiring loom from the cylinder head

Renew the fuel feed lines

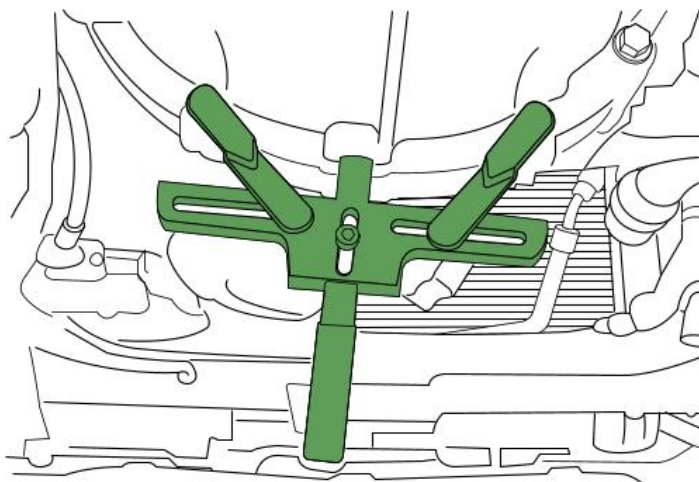
Remove the air filter housing and hoses

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Disconnect all the vacuum hoses from the cylinder head and ancillaries
Remove the cylinder head cover

Remove the spark plugs
Remove the ancillary drive belt
Remove the ancillary drive belt tensioner
Remove the underbody cover

Vehicles with manual gearbox:
Fit the crankshaft locking tool (11 8 180)

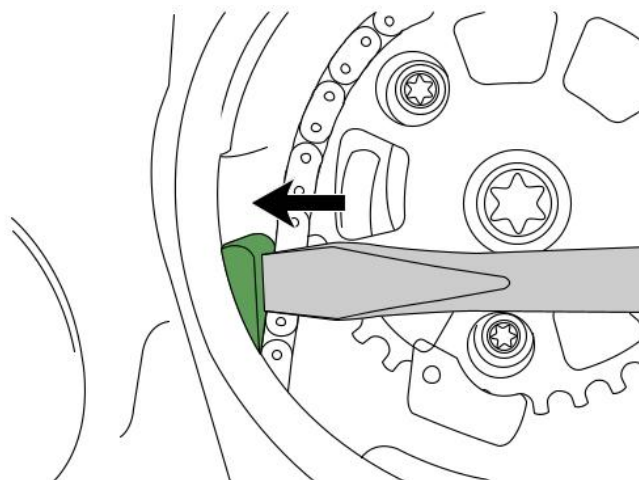


Vehicles with automatic transmission:
Fit the crankshaft locking tool (11 8 660)

All vehicles:
Remove the vibration damper
Note: Do not loosen the crankshaft gearwheel centre bolt

Vacuum pump:
Remove the sealing cap
Oil pump:
Remove the sealing cap

Loosen the chain tensioner piston



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Fit the tensioner locking pin

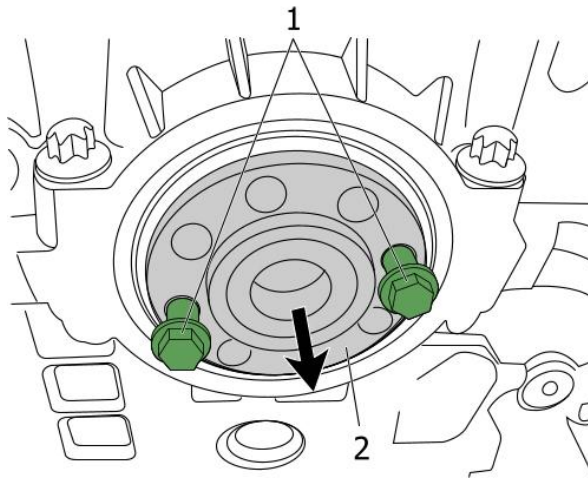
Lock the tensioner

Remove the crankshaft gearwheel centre bolt

Crankshaft pulley outer bolts:

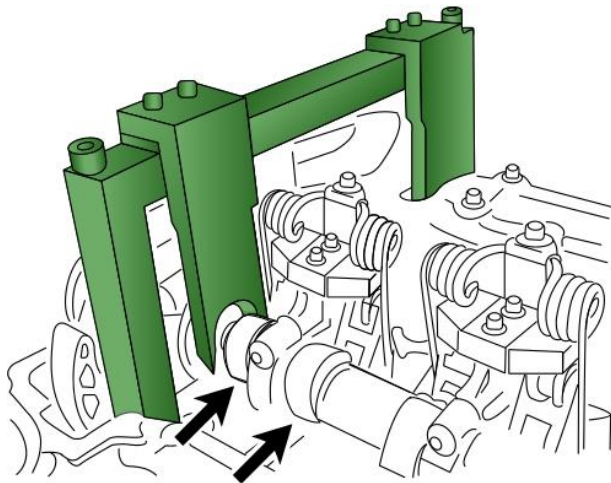
Refit the bolts shown (1)

Remove the crankshaft hub (2)



Remove the timing chain guide bolts from the cylinder head

Fit the camshaft locking tools (11 4 281, 11 4 282, 11 4 283)

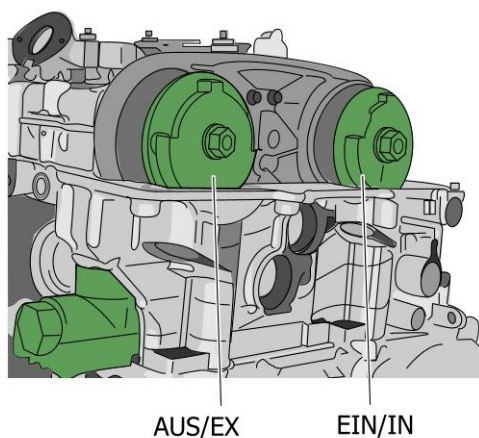


Remove the upper chain tensioner

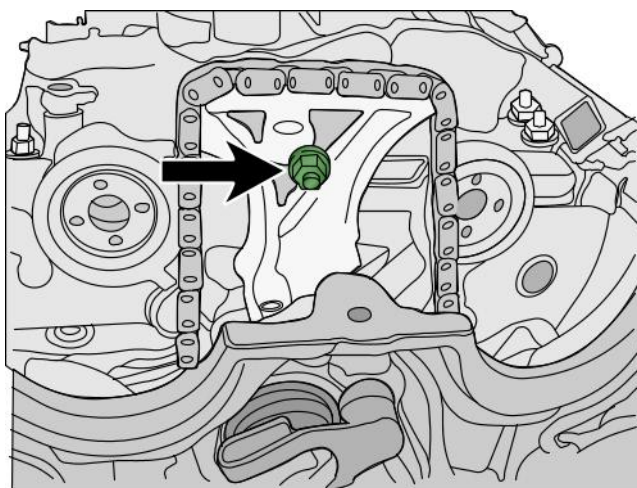
Remove the inlet camshaft timing adjuster (EIN/EI)

Remove the exhaust camshaft timing adjuster (AUS/EX)

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Remove the timing chain guide bolts from the cylinder head (x 2)

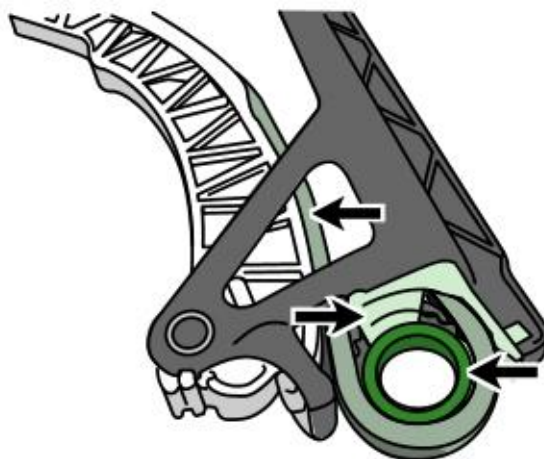


Remove the bearing pins for the chain guide rail and the tensioner rail

Remove the chain guide with the timing chain, guide rail and tensioner rail

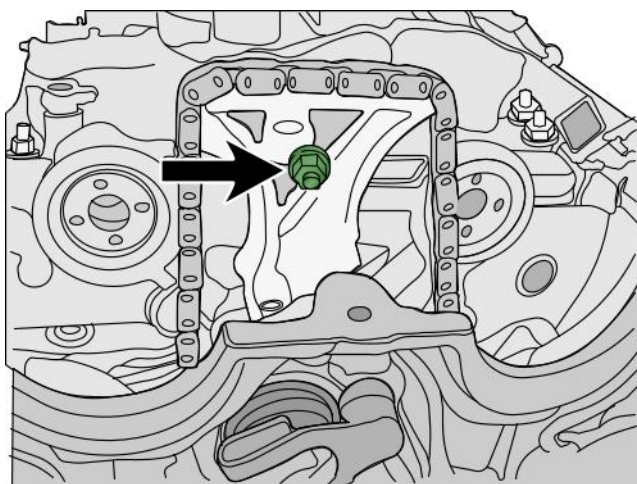
Installation

Refit the chain guide with the timing chain, guide rail and tensioner rail



Refit the timing chain guide bolts to the cylinder head (x 2)

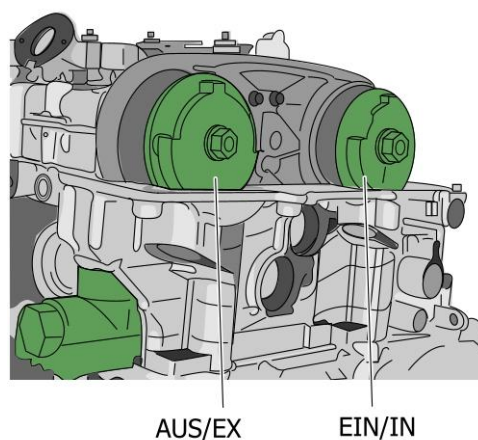
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Refit the crankshaft hub

Refit the exhaust camshaft adjuster (AUS/EX)

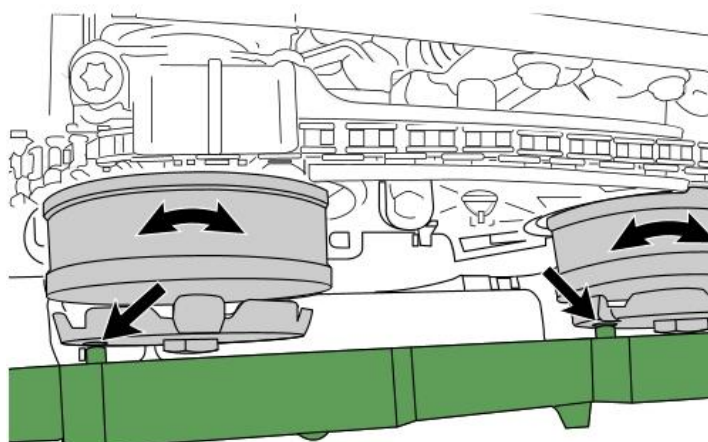
Refit the inlet camshaft timing adjuster (EIN/IN)



Tighten the camshaft gearwheel bolts lightly until there is no play

Fit the camshaft alignment tool (11 4 290)

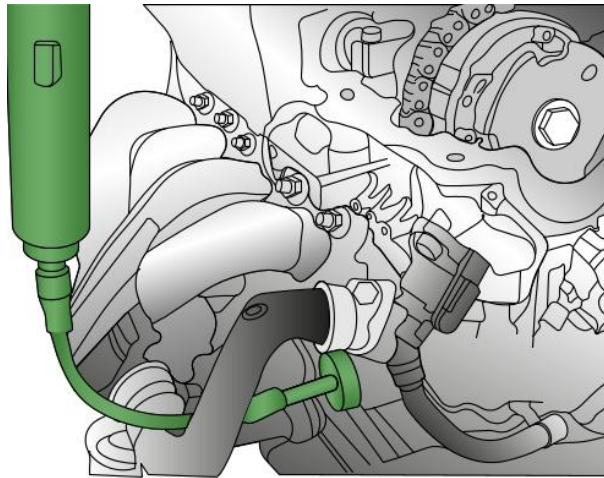
Rotate the camshaft sensor wheels until the retaining pins on the special tool fit correctly



Pre-tension the timing chain (0.6 Nm)

Use the special tools (00 9 250, 11 9 340)

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Tighten the camshaft gearwheel bolts

Installation is in the reverse order of removal

Torque settings

Timing chain guide

Timing chain tensioner

Crankshaft pulley

Crankshaft gearwheel

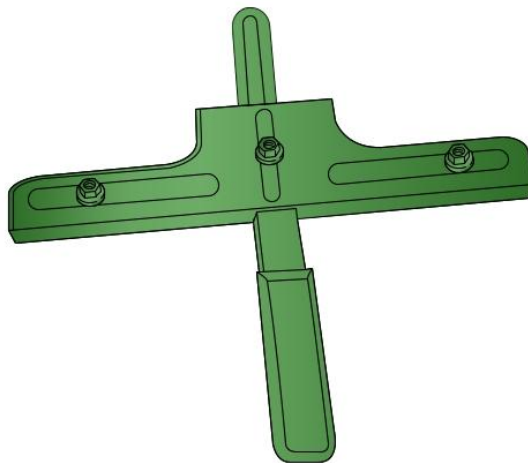
Valve cover

Spark plugs

Camshaft gearwheel

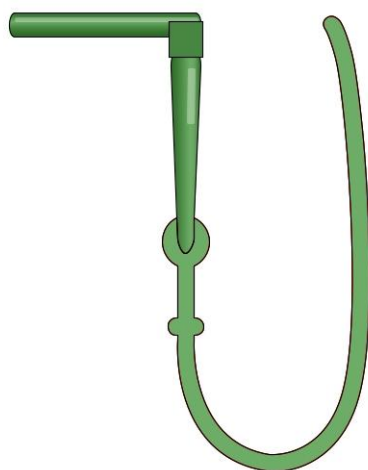
Special tools

Crankshaft locking tool: (11 8 180)

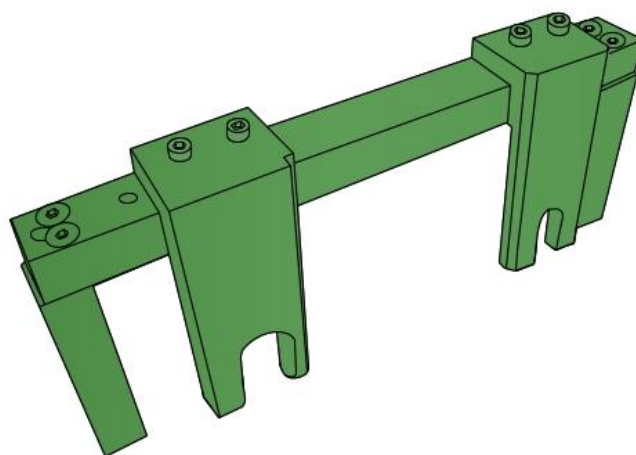


Crankshaft locking tool: (11 0 300)

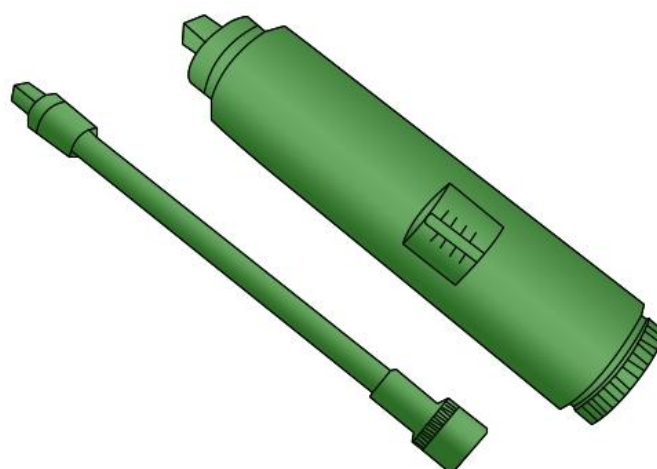
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Camshaft locking tool: (11 4 281, 11 4 282, 11 4 283)

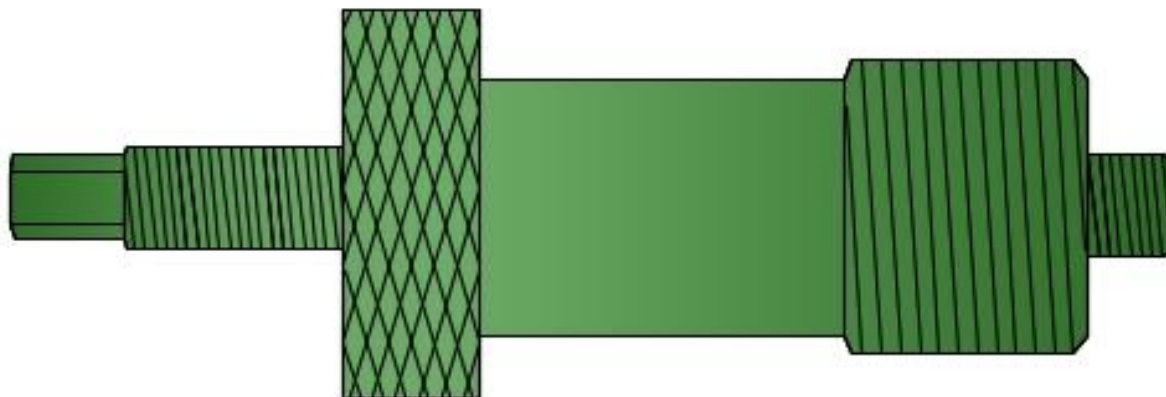


Adjustment tool: (00 9 250)

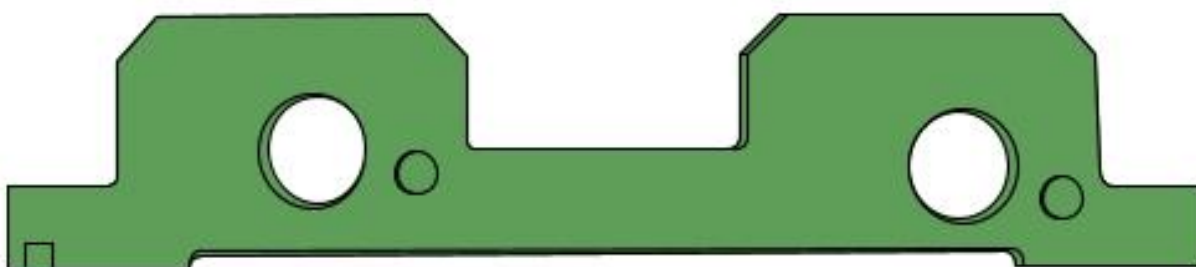


Adaptor: (11 9 340)

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Camshaft alignment tool: (11 4 290)



Flywheel locking tool: (11 8 660)

